Chapter 5

Relations

5.1 Basic Definitions

Definition: A ***relation*** is a set of ordered pairs. Specifically, a set R is a relation if $(\forall x)(x \in R \Rightarrow (\exists a)(\exists b)x = (a, b))$.

Example 5.1.1

(i) Let A be the set of all men; B be the set of all women.

The set of ordered pairs $M = \{x \in A \times B \mid x = (a, b), where\ a\ is\ the\ husband\ of\ b\}$ is a relation that definitions "*marriage.*"

(ii) Let A and B be the two sets. The Cartesian product $A \times B$ is perhaps the most obvious example of relations.

(iii) The empty set $\emptyset$ is a relation.

Proof: First, we shall prove $x \in \emptyset \Rightarrow (\exists a)(\exists b)x = (a, b)$

(Vacuous proof)

$x \in \emptyset \Leftrightarrow false$ (Lemma 3.5.7)

$\Rightarrow (x \in \emptyset \Rightarrow false) \wedge (false \Rightarrow x \in \emptyset)$ (E20)

$\Rightarrow (x \in \emptyset \Rightarrow false)$ (I2)

Hence, $x \in \emptyset \Rightarrow (\exists a)(\exists b)x = (a, b)$

$\Rightarrow (\forall x)(x \in \emptyset \Rightarrow (\exists a)(\exists b)x = (a, b))$ (Gen)

$\Rightarrow \emptyset$ is a relation. (Definition of relation) ■

(iv) (*The relation of equality*)

Let X be any set. The set $X_{=} = \{(a, b) \in X \times X \mid a = b\}$ is a relation.

(v) (*The relation of belonging*)

Let X be any set. The set $R_{\in} = \{(a, A) \in X \times \mathcal{P}(X) \mid a \in A\}$ is a relation.

(vi) Let X be any set. The set $R_{\subseteq} = \{(A, B) \in \mathcal{P}(X) \times \mathcal{P}(X) \mid A \subseteq B\}$ is a relation.

(vii) Let $\mathbf{R}$ be the set of all real numbers. The '*less than or equal*' relation between real numbers can be defined as a set of ordered pairs as follows: $R_{\leq} = \{(a, b) \in \mathbf{R} \times \mathbf{R} \mid a \leq b\}$

(viii) (*The congruence relation*)

Let m be a positive integer and $\mathbf{Z}$ be the set of all integers. The congruence relation (modulo m) is the relation:

$$\equiv_m = \{(x, y) \in \mathbf{Z} \times \mathbf{Z} \mid (\exists k)(k \in \mathbf{Z} \wedge x - y = km)\},$$

It can be shown that $(\exists k)(k \in \mathbf{Z} \wedge x - y = km)$ is equivalent to "the remainders of the two divisions x/m and y/m are identical."

(ix) (*The divisibility relation*)

Let $\mathbf{N}$ be the set of positive integers. The divisibility relation is the relation

$$\mid = \{(a, b) \in \mathbf{N} \times \mathbf{N} \mid (\exists k \in \mathbf{N})b = ak\}.$$

In other words, $(a, b) \in \mid$ iff a divides b. ■

Remark: Let R be a relation. Then $(a, b) \in R$ may be written as aRb. For instance, $(21, 36) \in \equiv_5$ is commonly written as $21 \equiv_5 36$ and $(3, 9) \in \mid$ is commonly written as $3 \mid 9$. Moreover, for the relation $X_=$, we usually write $a = b$ instead of $aX_=b$. Similarly, $aR_\in b$, $aR_\subseteq b$ and $aR_\leq b$ are conventionally written as $a \in b$, $a \subseteq b$ and $a \leq b$, respectively. Henceforth, we shall use $=$, $\in$, $\subseteq$ and $\leq$, respectively to denote the relations $X_=$, $R_\in$, $R_\subseteq$ and $R_\leq$, whenever is convenience.

Definition: Let R be a relation. The ***domain*** of R is the set $Dom(R) = \{x \mid (\exists y)(x, y) \in R\}$ and the ***range*** of R is the set $Ran(R) = \{y \mid (\exists x)(x, y) \in R\}$.

Dom(R) is the set consisting of the first coordinates of the ordered pairs in R. *Ran*(R) is the set consisting of the second coordinates of the ordered pairs in R.

Example 5.1.2

(i) Let $R = \{(a, \alpha), (b, \alpha), (a, \beta), (c, \alpha)\}$. $Dom(R) = \{a, b, c\}$ and $Ran(R) = \{\alpha, \beta\}$.

(ii) In the "*marriage*" relation M, *Dom*(M) is the set of all married men while *Ran*(M) is the set of all married women.

(iii) Let $R = X \times Y$. $Dom(R) = X$ and $Ran(R) = Y$.

(iv) In the relation $\emptyset$, $Dom(\emptyset) = Ran(\emptyset) = \emptyset$.

Proof: (Proof by Contradiction) Suppose $Dom(\emptyset) \neq \emptyset$. Then,

$(\exists x)x \in Dom(\emptyset)$ (Lemma 3.5.8)

$\Rightarrow a \in Dom(\emptyset)$ (EI)

$\Rightarrow (\exists y)(a, y) \in \emptyset$ (Definition of $Dom(\emptyset)$)

$\Rightarrow (a, b) \in \emptyset$ (EI)

$\Rightarrow (\exists z)z \in \emptyset$ (EQ)

$\Rightarrow \emptyset \neq \emptyset$ (Lemma 3.5.8)

But by Lemma ?(i), $\emptyset = \emptyset$. We thus have a contradiction!

Hence, $Dom(\emptyset) = \emptyset$.
$Ran(\emptyset) = \emptyset$ can be proved similarly. □

(v) For the relation of equality $X_=$, $Dom(X_=) = Ran(X_=) = X$.

(vi) For the relation of belonging $\in$, $Dom(\in) = X$ and $Ran(\in) = \mathcal{P}(X) - \{\emptyset\}$.

(vii) For the relation $\subseteq$, $Dom(\subseteq) = Ran(\subseteq) = \mathcal{P}(X)$.

(viii) For the relation $\leq$, $Dom(\leq) = Ran(\leq) = \mathbf{R}$.

(ix) For the relation $\equiv_m$, $Dom(\equiv_m) = Ran(\equiv_m) = \mathbf{Z}$.

(x) For the relation $|$, $Dom(|) = Ran(|) = \mathbf{N}$. ■

Let R be a relation, Then $R \subseteq Dom(R) \times Ran(R)$. Hence, any relation is a subset of a Cartesian product.

Definition: Let $R \subseteq X \times Y$, we say that R is a ***relation from*** X ***to*** Y. In particular, when $X = Y$, (i.e. $R \subseteq X \times X$), we say that R is a ***relation in*** X.

Lemma 5.1.1: Let R be a relation from X to Y. Then $Dom(R) \subseteq X$ and $Ran(R) \subseteq Y$.

Proof: Left as an exercise.

Example 5.1.3 In Example 5.1.1,

(i) The relation M is relation from the set of married men to the set of married women;
(ii) The relation $X \times Y$ is a relation from X to Y.
(iii) The relation $\emptyset$ is a relation from any set X to any set Y. This is because $\emptyset \subseteq X \times Y$ for any sets X and Y.
(iv) The relation $X_=$ is a relation in X
(v) The relation $\in$ is a relation from X to $\mathcal{P}(X)$.
(vi) The relation $\subseteq$ is a relation in $\mathcal{P}(X)$.
(vii) The relation $\leq$ is a relation in $\mathbf{R}$.
(viii) The congruence relation $\equiv_m$ is a relation in $\mathbf{Z}$.
(ix) The divisibility relation $|$ is a relation in $\mathbf{N}$. ■

5.2 Properties of Relation

Definition: Let R be a relation in a set X.

R is ***reflexive*** if $(\forall x \in X)(x, x) \in R$;
R is ***irreflexive*** if $(\forall x \in X)(x, x) \notin R$;
R is ***symmetric*** if $(x, y) \in R \Rightarrow (y, x) \in R$;
R is ***antisymmetric*** if $(x, y) \in R \wedge (y, x) \in R \Rightarrow x = y$;
R is ***asymmetric*** if $\forall x, y \in X, \sim ((x, y) \in R \wedge (y, x) \in R)$, or equivalently,

$$\forall x, y \in X, (x, y) \in R \Rightarrow (y, x) \notin R;$$

R is ***transitive*** if $(x, y) \in R \wedge (y, z) \in R \Rightarrow (x, z) \in R$.

Remark:

1. In the definition of reflexive property, $(\forall x \in X)(x,x) \in R$ is an abbreviation of $(\forall x)(x \in X \Rightarrow (x,x) \in R)$. (see Section 3.3.2)
2. In the definition of irreflexive property, $(\forall x \in X)(x,x) \notin R$ is an abbreviation of $(\forall x)(x \in X \Rightarrow (x,x) \notin R)$. (see Section 3.3.2)

Lemma 5.2.1: Let R be a relation in X. If $(\exists x \in X)(x,x) \in R$, then R is not asymmetric.

Proof: (Proof by contraction)

Suppose $(\exists x \in X)(x,x) \in R$ and R is asymmetric.

$(\exists x \in X)(x,x) \in R \Rightarrow a \in X \wedge (a,a) \in R$ (EI)

$\Rightarrow (a \in X \wedge a \in X) \wedge (a,a) \in R$ (E3)

$\Rightarrow \begin{cases} a \in X \wedge a \in X & \cdots (\mathrm{I}) \\ (a,a) \in R & \cdots (\mathrm{II}) \end{cases}$ (E9,I2)

R is asymmetric

$\Rightarrow \forall x, y \in X, \sim ((x,y) \in R \wedge (y,x) \in R)$ (Definition of asymmetric)

$\Rightarrow (\forall x)(\forall y)(x \in X \wedge y \in X \Rightarrow \sim ((x,y) \in R \wedge (y,x) \in R))$ (equivalent notation)

$\Rightarrow (a \in X \wedge a \in X \Rightarrow \sim ((a,a) \in R \wedge (a,a) \in R))$ (UI) . . . (III)

$\Rightarrow \sim ((a,a) \in R \wedge (a,a) \in R)$ ((I),(III),I3)

$\Rightarrow \sim (a,a) \in R$ (E3) . . . (IV)

$\Rightarrow (a,a) \in R \wedge \sim (a,a) \in R$ ((II),(IV),I6)

$\Rightarrow false$ (E1) ■

Remark: *Asymmetric* is not a synonyms of *not symmetric* as is illustrated in the following table.

Relation	Symmetric	Asymmetric
$\{(a,a),(a,b),(b,a)\}$	yes	no
$\{(a,a),(a,b)\}$	no	no
$\{(a,b)\}$	no	yes
$\emptyset$	yes	yes

Example 5.2.4

(i) The relation $\emptyset$ is a symmetric, antisymmetric, asymmetric and transitive relation. It is not reflexive unless it is considered as a relation in $\emptyset$. Note that $\emptyset \subseteq A \times A$ for *any* set A, including the case where $A = \emptyset$.

(ii) Let $X \neq \emptyset$. The relation $X \times X$ is a reflexive, symmetric and transitive relation in X. It is not antisymmetric unless X is a singleton.

(iii) Let $X \neq \emptyset$. The relation $=$ in X is a reflexive, symmetric and transitive relation in X. It is also antisymmetric. By Lemma 5.2.1, it is not asymmetric.

(iv) Let $X \neq \emptyset$. Owing to Lemma 3.2.3, the relation $\subseteq$ is a reflexive, antisymmetric and transitive relation in $\mathcal{P}(X)$. Again, by Lemma 5.2.1, it is not asymmetric. It is not symmetric because $\emptyset \subseteq X$ but $X \nsubseteq \emptyset$.

(v) From high school algebra, we learned that the relation $\leq$ is a reflexive, antisymmetric and transitive relation. It is neither symmetric nor asymmetric.

(vi) The congruence relation (modulo m), $\equiv_m = \{(x,y) \in \mathbf{Z} \times \mathbf{Z} \mid (\exists k)(k \in \mathbf{Z} \wedge x - y = km)\}$, is a reflexive, symmetric and transitive relation.

Proof: (i) $\equiv_m$ is reflexive: (Direct proof)

Let $x \in \mathbf{Z}$. Then $x \in \mathbf{Z} \wedge x \in \mathbf{Z}$ (E3)

$\Rightarrow (x,x) \in \mathbf{Z} \times \mathbf{Z}$ (Definition of $\times$) . . . (I)

Moreover, $x - x = 0 = 0 \cdot m$ and $0 \in \mathbf{Z}$.

$\Rightarrow 0 \in \mathbf{Z} \wedge x - x = 0 \cdot m$ (I6)

$\Rightarrow (\exists k)(k \in \mathbf{Z} \wedge x - x = k \cdot m)$ (EQ) . . . (II)

$\Rightarrow (x,x) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)(k \in \mathbf{Z} \wedge x - x = k \cdot m)$ ((I),(II),I6)

$\Rightarrow x \equiv_m x$ (Definition of $\equiv_m$).

(ii) $\equiv_m$ is symmetric: (Direct proof)

Let $x, y \in \mathbf{Z}$. Then $x \equiv_m y$

$\Rightarrow (x,y) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)\,(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ (Definition of $\equiv_m$)

$\Rightarrow (x,y) \in \mathbf{Z} \times \mathbf{Z}$ and $(\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ (I2,E9)

$\Rightarrow (x \in \mathbf{Z}$ and $y \in \mathbf{Z})$ and $(\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ (Definition of $\times$)

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z}$ and $(\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ (Definition of $\times$)

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z}$ and $(a \in \mathbf{Z} \wedge x - y = a \cdot m)$ (EI)

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z}$ and $(-a \in \mathbf{Z} \wedge x - y = a \cdot m)$ $(a \in \mathbf{Z} \Leftrightarrow -a \in \mathbf{Z})$

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z}$ and $(-a \in \mathbf{Z} \wedge y - x = (-a) \cdot m)$ $(p = q \Leftrightarrow -p = -q)$

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z}$ and $(\exists k)(k \in \mathbf{Z} \wedge y - x = k \cdot m)$ (EQ)

$\Rightarrow (y,x) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)(k \in \mathbf{Z} \wedge y - x = k \cdot m)$ (I6)

$\Rightarrow y \equiv_m x$. (Definition of $\equiv_m$)

(iii) $\equiv_m$ is transitive: (Direct proof)

Let $x, y, z \in \mathbf{Z}$. Then $x \equiv_m y$ and $y \equiv_m z$

$\Rightarrow (x,y) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ and

$(y,z) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)(k \in \mathbf{Z} \wedge y - z = k \cdot m)$ (Definition of $\equiv_m$)

$\Rightarrow (x \in \mathbf{Z} \wedge y \in \mathbf{Z}) \wedge (\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ and

$(y \in \mathbf{Z} \wedge z \in \mathbf{Z}) \wedge (\exists k)(k \in \mathbf{Z} \wedge y - z = k \cdot m)$ (Definition of $\times$)

$\Rightarrow x \in \mathbf{Z}$ and $z \in \mathbf{Z}$ and

$(\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ and $(\exists k)(k \in \mathbf{Z} \wedge y - z = k \cdot m)$ (E9, I2)

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and

$(\exists k)(k \in \mathbf{Z} \wedge x - y = k \cdot m)$ and $(\exists k)(k \in \mathbf{Z} \wedge y - z = k \cdot m)$ (Definition of $\times$)

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and

$(a \in \mathbf{Z} \wedge x - y = a \cdot m)$ and $(b \in \mathbf{Z} \wedge y - z = b \cdot m)$ (EI)

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and

$a \in \mathbf{Z}$ and $(x - y = a \cdot m)$ and $b \in \mathbf{Z}$ and $(y - z = b \cdot m)$ (E9,I2)

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and $a \in \mathbf{Z}$ and $b \in \mathbf{Z}$ and

$(x - y) + (y - z) = a \cdot m + b \cdot m$ $(a = b, c = d \Rightarrow a + c = b + d)$

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and $a \in \mathbf{Z}$ and $b \in \mathbf{Z}$ and $x - z = (a + b) \cdot m$

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and $(a + b) \in \mathbf{Z}$ and $x - z = (a + b) \cdot m$ $(a, b \in \mathbf{Z} \Rightarrow (a + b) \in \mathbf{Z})$

$\Rightarrow (x,z) \in \mathbf{Z} \times \mathbf{Z}$ and $(a + b) \in \mathbf{Z} \wedge x - z = (a + b) \cdot m$ (I6)

$\Rightarrow (x, z) \in \mathbf{Z} \times \mathbf{Z}$ and $(\exists k)(k \in \mathbf{Z} \wedge x - z = k \cdot m)$ (EQ)

$\Rightarrow (x, z) \in \mathbf{Z} \times \mathbf{Z} \wedge (\exists k)(k \in \mathbf{Z} \wedge x - z = k \cdot m)$ (I6)

$\Rightarrow x \equiv_m z$. (Definition of $\equiv_m$) $\square$

$\equiv_m$ is *not* antisymmetric because $m \equiv_m 2m$ and $2m \equiv_m m$ but $m \neq 2m$. It is also *not* asymmetric.

(vii) The divisibility relation $|$ is reflexive, antisymmetric and transitive.

Proof: (i) (Direct proof)

Let $x \in \mathbf{N}$.

Then $1 \in \mathbf{N}$ and $x = x \cdot 1$

$\Rightarrow (1 \in \mathbf{N}) \wedge (x = x \cdot 1)$ (I6)

$\Rightarrow (\exists k \in \mathbf{N})(x = xk)$ (EQ)

$\Rightarrow x \mid x$ (Definition of $|$)

Therefore, $(\forall x \in \mathbf{N})(x, x) \in | \Rightarrow |$ is reflexive.

(ii) (Direct proof)

Suppose $a \mid b$ and $b \mid a$.

Then, $(\exists k \in \mathbf{N})(b = ak)$ and $(\exists k' \in \mathbf{N})(a = bk')$. (Definition of $|$)

$\Rightarrow ((m \in \mathbf{N}) \wedge (b = am))$ and $((n \in \mathbf{N}) \wedge (a = bn))$ (EI)

$\Rightarrow (m \in \mathbf{N})$ and $(b = am)$ and $(n \in \mathbf{N})$ and $(a = bn)$ (E9, I2)

$\Rightarrow (b = (bn)m)$ ($\text{Sub}_=$)

$\Rightarrow (b = b(nm))$ (multiplication is associative)

$\Rightarrow mn = 1$

Since $m, n \in \mathbf{N}$, it follows that $(mn = 1) \Rightarrow (m = 1) \wedge (n = 1)$

As a result, $a = bn \Rightarrow a = b \cdot 1 \Rightarrow a = b$.

Hence, $|$ is antisymmetric.

(iii) (Direct proof)

Let $a \mid b$ and $b \mid c$.

Then, $(\exists k \in \mathbf{N})(b = ak)$ and $(\exists k \in \mathbf{N})(c = bk)$. (Definition of $|$)

$\Rightarrow (m \in \mathbf{N}) \wedge (b = am)$ and $(n \in \mathbf{N}) \wedge (c = bn)$ (EI)

$\Rightarrow (m \in \mathbf{N})$ and $(b = am)$ and $(n \in \mathbf{N})$ and $(c = bn)$ (E9, I2)

$\Rightarrow c = (am)n$ ($\text{Sub}_=$)

$\Rightarrow c = a(mn)$ (multiplication is associative) . . . (I)

Since $m, n \in \mathbf{N} \Rightarrow m \cdot n \in \mathbf{N}$, . . . (II)

we thus have: $mn \in \mathbf{N} \wedge c = a(mn)$ ((I),(II), I6)

$\Rightarrow (\exists K \in \mathbf{N})(c = aK)$ (EQ)

$\Rightarrow a \mid c$ (Definition of $|$)

Hence, $|$ is transitive.

$|$ is not symmetric because $3 \mid 9$ but $9 \nmid 3$ (note: this is a counterexample).

$|$ is also not asymmetric; a counterexample is $3 \mid 3$. ■

5.3 Equivalence Relations

Definition: A relation R in a set X is an ***equivalence relation*** if R is reflexive, symmetric and transitive.

Example 5.3.5 From Example 5.2.4, we immediately obtain the following results:
(i) The relation $X \times X$ is an equivalence relation in X, for any set X.
(ii) The relation of equality $=$ is an equivalence relation in X, for any set X.
(iii) The congruence relation $\equiv_m$ is an equivalence relation in $\mathbf{Z}$. ■

Definition: Let R be an equivalence relation in a set X. For each $x \in X$, the ***equivalence class of x w.r.t. R*** is the set $[x]/R = \{y \mid y \in X \wedge (x, y) \in R\} = \{y \in X \mid (x, y) \in R\}$.

In other words, the equivalence class of x w.r.t. R consists of all the elements of X which form an ordered pair with x in R, and in each of such ordered pairs, x is the first coordinate.

Since $(x, y) \in R$ and $R \subseteq X \times X$, it is easily verified that $(x, y) \in R \Leftrightarrow (y \in X \wedge (x, y) \in R)$. It follows that $[x]/R$ can be defined as:

$$[x]/R \quad = \quad \{y \mid (x, y) \in R\}$$

Example 5.3.6

(i) Let X be any set and $R = X \times X$. $\forall x \in X$, $[x]/R = X$.

Proof: (Direct proof) Let $R = X \times X$. . . . (I)

We are to prove that $(\forall x)(x \in X \Rightarrow [x]/R = X)$.

(Direct proof) Let $x \in X$. . . . (II)

We shall first prove that $y \in [x]/R \Leftrightarrow y \in X$.

(Bidirectional proof)

$\Leftarrow$) Let $y \in X$. . . . (III)

Then $x \in X \wedge y \in X$ ((II),(III),I6)

$\Rightarrow (x, y) \in X \times X$ (Definition of $X \times X$)

$\Rightarrow (x, y) \in R$ ((I), Sub$_=$)

$\Rightarrow y \in [x]/R$ (Definition of $[x]/R$)

$\Rightarrow$) Let $y \in [x]/R$.

Then $y \in X \wedge (x, y) \in R$ (Definition of $[x]/R$)

$\Rightarrow y \in X$ (I2)

We thus have $y \in [x]/R \Leftrightarrow y \in X$

$\Rightarrow (\forall y)(y \in [x]/R \Leftrightarrow y \in X)$ (Gen)

$\Rightarrow [x]/R = X$. (Principle of Extension)

Hence, $x \in X \Rightarrow [x]/R = X$

$\Rightarrow (\forall x)(x \in X \Rightarrow [x]/R = X)$. (Gen) □

(ii) Consider the equivalence relation $=$ in a set X. $\forall x \in X$, $[x]/= = \{x\}$.

Proof: (Direct proof) Let $x \in X$.

$y \in \{x\} \Leftrightarrow y = x$ (Definition of $\{x\}$)

$\Leftrightarrow x = y$ (= is symmetric)

$\Leftrightarrow (x, y) \in =$ (Definition of =)

$\Leftrightarrow y \in [x]/=$ (Definition of $[x]/=$)

Hence, $y \in [x]/= \Leftrightarrow y \in \{x\}$

$\Rightarrow (\forall y)(y \in [x]/= \Leftrightarrow y \in \{x\})$ (Gen)

$\Rightarrow [x]/= = \{x\}$. (Principle of Extension) □

(iii) Consider the congruence relation $\equiv_m$.

For each $x \in \mathbf{Z}$, $x = lm + r_x$ for some $l, r_x \in \mathbf{Z}$ where $0 \leq r_x < m$ (note that r_x is the remainder of dividing x by m)[3]. Then,

$$[x]/\equiv_m = \{y \in \mathbf{Z} \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\},$$

Proof: Let $z \in [x]/\equiv_m$.

Then $(x, z) \in \equiv_m \Rightarrow (\exists k \in \mathbf{Z})(x - z = km)$ (Definition of $\equiv_m$)

$\Rightarrow x - z = am$, *where* $a \in \mathbf{Z}$ (EI)

$\Rightarrow (lm + r_x) - z = am$, for some $l \in$ Z

($\because x = lm + r_x$ for some $l \in$ Z)

$\Rightarrow (\alpha m + r_x) - z = am$, *where* $\alpha \in \mathbf{Z}$ (EI)

$\Rightarrow z = (\alpha - a)m + r_x$

$\Rightarrow z = bm + r_x$, *where* $b = (\alpha - a) \in \mathbf{Z}$

$\Rightarrow (\exists q \in \mathbf{Z})(z = qm + r_x)$

$\Rightarrow z \in \{y \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\})$

Let $z \in \{y \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\}$.

Then $(\exists q \in \mathbf{Z})(z = (qm + r_x))$

$\Rightarrow (\exists q \in \mathbf{Z})(z - x = (qm + r_x) - x)$

$\Rightarrow (\exists q, l \in \mathbf{Z})(z - x = (qm + r_x) - (lm + r_x))$ ($x = lm + r_x$ for some $l \in$ Z)

$\Rightarrow (\exists q, l \in \mathbf{Z})(z - x = qm - lm)$

$\Rightarrow (\exists q, l \in \mathbf{Z})(x - z = (l - q)m)$

$\Rightarrow (x - z = (a - b)m)$, *where* $a, b \in \mathbf{Z}$ (EI)

$\Rightarrow (x - z = cm)$, *where* $c = (a - b) \in \mathbf{Z}$

$\Rightarrow (\exists q' \in \mathbf{Z})(x - z = q'm)$ (EQ)

$\Rightarrow (x, z) \in \equiv_m$ (Definition of $\equiv_m$)

$\Rightarrow z \in [x]/\equiv_m$ (Definition of $[x]/\equiv_m$)

Therefore, $z \in [x]/\equiv_m \Leftrightarrow z \in \{y \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\}$

$\Rightarrow (\forall z)(z \in [x]/\equiv_m \Leftrightarrow z \in \{y \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\})$ (Gen)

$\Rightarrow [x]/\equiv_m = \{y \mid (\exists q \in \mathbf{Z})(y = qm + r_x)\}$ (Principle of Extension)

Remark: The defining property $(\exists q \in \mathbf{Z})(y = qm + r_x)$ states that the remainder of the division y/m is r_x which is the remainder of the division x/m. Therefore, $[x]/\equiv_m$ consists of all the integers whose divisions by m produce the same remainder r_x. ■

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3 The existence of l and r_x is guaranteed by Elementary Number Theory.

Definition: Let R be an equivalence relation in a set X. ***The collection of equivalence classes w.r.t. R*** is the set

$$[X]/R \;=\; \{S \mid (\exists x)(x \in X \wedge S = [x]/R)\} \;=\; \{[x]/R \mid x \in X\}$$

Example 5.3.7 Consider the equivalence relations in Example 5.3.5.

(i) Since $[x]/(X \times X) = X, \forall x \in X$, therefore $[X]/(X \times X) = \{[x]/(X \times X) \mid x \in X\} = \{X\}$. Note that the parentheses enclosing $X \times X$ is included for clarity.

(ii) Since $[x]/= \;= \{x\}, \forall x \in X$, therefore $[X]/= \;= \{[x]/= \mid x \in X\} = \{\{x\} \mid x \in X\}$

(iii) Let $x, y \in \mathbf{Z}$. Then $x = qm + r_x$ and $y = q'm + r_y$ for some $q, q', r_x, r_y \in \mathbf{Z}$ where $0 \leq r_x, r_y < m$.

If $r_x = r_y$, then $[x]/\equiv_m = \{z \in \mathbf{Z} \mid (\exists q \in \mathbf{Z})(z = qm + r_x)\}$ (Example 5.3.6(iii))

$= \{z \in \mathbf{Z} \mid (\exists q \in \mathbf{Z})(z = qm + r_y)\}$ $(r_x = r_y)$

$= [y]/\equiv_m$. (Example 5.3.6(iii))

If $r_x \neq r_y$ (without loss of generality, we assume then $r_x > r_y$), then as $0 \leq r_x, r_y < m$, we have $0 < r_x - r_y < m$. This implies that:

$$\begin{aligned}
& qm + 0 < qm + (r_x - r_y) < qm + m \\
\Rightarrow\; & qm < qm + (r_x - r_y) < (q+1)m \\
\Rightarrow\; & qm + r_y < qm + (r_x - r_y) + r_y < (q+1)m + r_y \\
\Rightarrow\; & qm + r_y < qm + r_x < (q+1)m + r_y \\
\Rightarrow\; & qm + r_y < x < (q+1)m + r_y \\
\Rightarrow\; & (\nexists Q \in \mathbf{Z})x = Qm + r_y \\
\Rightarrow\; & x \notin [y]/\equiv_m \qquad \text{(Definition of } [y]/\equiv_m\text{)}
\end{aligned}$$

But $x \in [x]/\equiv_m$ $(\because (x,x) \in \equiv_m)$

We thus have $\Big(x \in [x]/\equiv_m\Big) \wedge \Big(x \notin [y]/\equiv_m\Big)$

$$\begin{aligned}
\Rightarrow\; & (\exists z)\Big(\Big(z \in [x]/\equiv_m\Big) \wedge \Big(z \notin [y]/\equiv_m\Big)\Big) && \text{(EQ)} \\
\Rightarrow\; & (\exists z)\Big(\Big(z \in [x]/\equiv_m\Big) \wedge \Big(z \notin [y]/\equiv_m\Big)\Big) \vee (\exists z)\Big(\Big(z \notin [x]/\equiv_m\Big) \wedge \Big(z \in [y]/\equiv_m\Big)\Big) && \text{(I1)} \\
\Rightarrow\; & [x]/\equiv_m \neq [y]/\equiv_m . && \text{(Lemma 3.2.2)}
\end{aligned}$$

Hence, $[x]/\equiv_m = [y]/\equiv_m$ iff $r_x = r_y$.

Now, as the remainder obtained from dividing any integer by m is in the range from 0 to $m-1$, they are *exactly m* distinct remainders resulting in *exactly m* distinct equivalence classes in $[\mathbf{Z}]/\equiv_m$. As a result,

$$[\mathbf{Z}]/\equiv_m = \Big\{[r]/\equiv_m \mid 0 \leq r < m\Big\}$$

For instance, when m=3, as all the multiples of 3 have the same remainder, namely 0, when dividing by 3, their equivalence classes are identical. We thus have

$$\cdots = \Big([-3k]/\equiv_3\Big) = \cdots = \Big([-6]/\equiv_3\Big) = \Big([-3]/\equiv_3\Big) = \Big([0]/\equiv_3\Big) = \Big([3]/\equiv_3\Big) = \Big([6]/\equiv_3\Big) = \cdots = \Big([3k]/\equiv_3\Big)\cdots$$

By a similar argument, we have

$$\cdots = \Big([3(-k)+1]/\equiv_3\Big) = \cdots = \Big([-5]/\equiv_3\Big) = \Big([-2]/\equiv_3\Big) = \Big([1]/\equiv_3\Big) = \Big([4]/\equiv_3\Big) = \Big([7]/\equiv_3\Big) = \cdots = \Big([3k+1]/\equiv_3\Big)\cdots$$

$$\cdots = \Big([3(-k)+2]/\equiv_3\Big) = \cdots = \Big([-4]/\equiv_3\Big) = \Big([-1]/\equiv_3\Big) = \Big([2]/\equiv_3\Big) = \Big([5]/\equiv_3\Big) = \Big([8]/\equiv_3\Big) = \cdots = \Big([3k+2]/\equiv_3\Big)\cdots$$

Hence, $[\mathbf{Z}]/\equiv_3 = \Big\{[0]/\equiv_3, [1]/\equiv_3, [2]/\equiv_3\Big\}$ ■

Definition: Let X be a set. A collection of sets C is a ***partition*** of X if

(i) $\bigcup_{S \in C} S = X$, and

(ii) $\forall S \in C, S \neq \emptyset$, and

(iii) $\forall S, S' \in C, S \neq S' \Rightarrow S \cap S' = \emptyset$.

Remark: Condition (ii) is equivalent to $(\forall S)(S \in C \Rightarrow S \neq \emptyset)$, and Condition (iii) is equivalent to $(\forall S)(\forall S')\big(S \in C \wedge S' \in C \Rightarrow (S \neq S' \Rightarrow S \cap S' = \emptyset)\big)$. (See Section 3.3.2)

Example 5.3.8 Consider the equivalence relations in Example 5.3.5. We want to show that the collection of equivalence classes for each of them is a partition of the set in which the relation is defined.

(i) For the equivalence relation $X \times X$, we assume $X \neq \emptyset$.

In Example 5.3.7(i) we showed that $[X]/(X \times X) = \{X\}$.

a. By Example 4.6.6(3), we have $\bigcup_{S \in [X]/(X \times X)} S = \bigcup_{S \in \{X\}} S = X$.

b. Since $[X]/(X \times X) = \{X\}$, we have $\forall S \in [X]/(X \times X), S = X$. But $X \neq \emptyset$ by assumption. Therefore, $S \neq \emptyset$.

c. We are to prove that: $\forall S, S' \in [X]/(X \times X), (S \neq S' \Rightarrow S \cap S' = \emptyset)$.

Let $S, S' \in [X]/(X \times X)$. As $[X]/(X \times X) = \{X\}$, we have $(S = X)$ and $(S' = X)$ which implies that $S = S'$.

Therefore, $S \neq S'$ is false, $\forall S, S' \in [X]/(X \times X)$.

By the vacuous proof method, $\forall S, S' \in [X]/(X \times X), (S \neq S' \Rightarrow S \cap S' = \emptyset)$ thus holds true.

In conclusion, $[X]/(X \times X) = \{X\}$ is a partition of X.

(ii) For the equivalence relation $=$, we have shown in Example 5.3.7(ii) that $[X]/= = \{\{x\} \mid x \in X\}$.

a. Let $x \in X$. Since $x \in \{x\}$ and $\{x\} \in [X]/=$, we have $x \in S$, for some $S \in [X]/=$. (EQ)

Therefore $x \in \bigcup_{S \in [X]/=} S$. (Definition of $\bigcup$)

On the other hand, if $x \in \bigcup_{S \in [X]/=} S$, then $x \in S$ for some $S \in [X]/=$ (Definition of $\bigcup$).

This implies that $x \in \{y\}$ for some $y \in X$ ($\because [X]/== \{\{x\} \mid x \in X\}$).

But then $x \in \{y\}$ implies $x = y$. Hence, $x \in X$.

Consequently, $x \in \bigcup_{S \in [X]/=} S \Leftrightarrow x \in X$

$$\Rightarrow (\forall x)\left(x \in \bigcup_{S \in [X]/=} S \Leftrightarrow x \in X\right) \quad \text{(Gen)}$$

$$\Rightarrow \bigcup_{S \in [X]/=} S = X. \quad \text{(Principle of Extension)}$$

b. Since $\forall x \in X, ([x]/=) = \{x\} \neq \emptyset$, therefore $\forall S \in [X]/=, S \neq \emptyset$.

c. Let $S, S' \in [X]/=$ such that $S \cap S' \neq \emptyset$.

Since $[X]/= = \{\{x\} \mid x \in X\}$,

$S, S' \in [X]/= \Rightarrow S, S' \in \{\{x\} \mid x \in X\}$

$\Rightarrow (\exists x \in X)S = \{x\}$ and $(\exists y \in X)S' = \{y\}$

$\Rightarrow S = \{a\}$ and $S' = \{b\}$, where $a, b \in X$ (EI)

As $S \cap S' \neq \emptyset$, we have $\{a\} \cap \{b\} \neq \emptyset$.

It follows from Lemma 3.5.8 that $(\exists z)z \in (\{a\} \cap \{b\})$

$\Rightarrow k \in (\{a\} \cap \{b\})$ (EI)

$\Rightarrow (k \in \{a\}) \wedge (k \in \{b\})$ (Definition of $\cap$)

$\Rightarrow (k = a) \wedge (k = b)$ (Definitions of $\{a\}$ and $\{b\}$)

$\Rightarrow a = b$ (= is symmetric and transitive, or $\text{Sub}_=$)

$\Rightarrow \{a\} = \{b\}$ (Lemma 3.2.1(i), UI, $\text{Sub}_=$)

$\Rightarrow S = S'$ ($S = \{a\}$ and $S' = \{b\}$, $\text{Sub}_=$)

In conclusion, $[X]/=$ is a partition of X.

(iii) For the equivalence relation $\equiv_m$, we have shown in Example 5.3.7(iii) that $[\mathbf{Z}]/\equiv_m = \left\{[r]/\equiv_m \mid 0 \leq r < m\right\}$ ... (I).

a. $y \in \bigcup_{S \in [\mathbf{Z}]/\equiv_m} S \Leftrightarrow y \in S$, for some $S \in [\mathbf{Z}]/\equiv_m$ (Definition of $\bigcup$)

$\Leftrightarrow y \in [k]/\equiv_m, 0 \leq k < m$, where $k \in \mathbf{Z}$ ($\because$ (I))

$\Leftrightarrow y \in \{z \in \mathbf{Z} \mid (\exists q \in \mathbf{Z})(z = qm + r_k)\}$ (Example 5.3.6(iii))

$\Leftrightarrow y \in \{z \in \mathbf{Z} \mid (\exists q \in \mathbf{Z})(z = qm + k)\}$ ($0 \leq k < m \Rightarrow k = r_k$)

$\Leftrightarrow (\exists q \in \mathbf{Z})(y = qm + k)$, where $k, q \in \mathbf{Z}, 0 \leq k < m$

$\Leftrightarrow y = \alpha m + k$, *where* $k, \alpha \in \mathbf{Z}, 0 \leq k < m$ ($\Rightarrow$) EI, $\Leftarrow$) EQ)

$\Leftrightarrow y \in \mathbf{Z}$ (Elementary Number Theory)

Therefore, $\bigcup_{S \in [\mathbf{Z}]/\equiv_m} S = \mathbf{Z}$.

b. $\forall S \in [\mathbf{Z}]/\equiv_m$, $S = [r]/\equiv_m$, for some $0 \leq r < m$. Since $r = 0 \cdot m + r$ and $0 \in \mathbf{Z}$, therefore, $r \in [r]/\equiv_m$ implying $S \neq \emptyset$.

c. Let $S, S' \in [\mathbf{Z}]/\equiv_m$ such that $S \cap S' \neq \emptyset$.

Since $[\mathbf{Z}]/\equiv_m = \left\{[r]/\equiv_m \mid 0 \leq r < m\right\}$,

$S, S' \in [\mathbf{Z}]/\equiv_m \Rightarrow (\exists r \in \mathbf{Z})S = [r]/\equiv_m$ and $(\exists r' \in \mathbf{Z})S' = [r']/\equiv_m$, where $0 \leq r, r' < m$

$\Rightarrow S = [a]/\equiv_m$ and $S' = [b]/\equiv_m$, where $0 \leq a, b < m$

As $S \cap S' \neq \emptyset$, we have $[a]/\equiv_m \cap [b]/\equiv_m \neq \emptyset$.

It implies that $(\exists z)z \in \left([a]/\equiv_m \cap [b]/\equiv_m\right)$ (Lemma 3.5.8)

$\Rightarrow k \in \left([a]/\equiv_m \cap [b]/\equiv_m\right)$, where $k \in \mathbf{Z}$ (EI)

$\Rightarrow k \in [a]/\equiv_m \wedge k \in [b]/\equiv_m$ (Definition of $\cap$)

$\Rightarrow (k = qm + a) \wedge (k = q'm + b)$, where $q, q' \in \mathbf{Z}$

(Definition of $[a]/\equiv_m$ and $[b]/\equiv_m$)

$\Rightarrow (qm + a) = (q'm + b)$ ($\text{Sub}_=$)

$\Rightarrow a - b = (q' - q)m$

Suppose $q \neq q'$. Without loss of generality, we assume $q' > q$.

Since $q, q' \in \mathbf{Z}$, $q' > q \Rightarrow q' - q \geq 1 \Rightarrow (q' - q)m \geq m$.

But $0 \leq a, b < m \Rightarrow |a - b| < m$.

We thus have $m > |a - b| = (q' - q)m \geq m$.

$\Rightarrow m > m$, a contradiction!

Hence, $q = q'$.

It follows that $a - b = (q' - q)m = 0 \Rightarrow a = b$.

Consequently, $S = \left([a]/\equiv_m\right) = \left([b]/\equiv_m\right) = S'$.

In conclusion, $[\mathbf{Z}]/\equiv_m$ is a partition of $\mathbf{Z}$. ■

In each of the above three examples, we showed that the collection of the equivalence classes forms a partition of the set in which the equivalence relation is defined. The following lemma shows that this is not a coincidence. Before presenting the lemma and its proof, we shall write down a few lemmas that are useful in shortening our proofs in the following remark.

Remark: (***Some common practices***)

CP 5.3(a): *Let R be a reflexive relation in X and $a \in X$, then $(a, a) \in R$.*

Proof: (Direct proof) Let R be a reflexive relation in X and $a \in X$. . . . (I)

Then R is reflexive $\Rightarrow (\forall x)(x \in X \Rightarrow (x, x) \in R)$ (Definition of Reflexivity)

$\Rightarrow (a \in X \Rightarrow (a, a) \in R)$ (UI) . . . (II)

$\Rightarrow (a, a) \in R$ ((I),(II),I3) □

CP 5.3(b): *Let R be a symmetric relation in X and $(a, b) \in R$, then $(b, a) \in R$.*

Proof: (Direct proof) Let R be a symmetric relation in X and $(a, b) \in R$. . . . (I)

Then R is symmetric

$\Rightarrow (\forall x)(\forall y)((x, y) \in R \Rightarrow (y, x) \in R)$ (Definition of symmetricity)

$\Rightarrow (a, b) \in R \Rightarrow (b, a) \in R$ (UI) . . . (II)

$\Rightarrow (b, a) \in R$ ((I),(II),I3) □

CP 5.3(c): *Let R be a antisymmetric relation in X, and $(a, b) \in R \wedge (b, a) \in R$, then $a = b$.*

Proof: (Direct proof) Let R be an antisymmetric relation in X, and $(a, b) \in R \wedge (b, a) \in R$. . . . (I)

Then R is antisymmetric

$\Rightarrow (\forall x)(\forall y)((x, y) \in R \wedge (y, x) \in R \Rightarrow x = y)$ (Definition of antisymmetricity)

$\Rightarrow ((a, b) \in R \wedge (b, a) \in R \Rightarrow a = b)$ (UI) . . . (II)

$\Rightarrow a = b$ ((I),(II),I3) □

CP 5.3(d): *Let R be a transitive relation in X, and $(a, b) \in R \wedge (b, c) \in R$, then $(a, c) \in R$.*

Proof: (Direct proof) Let R be a transitive relation in X, and $(a, b) \in R \wedge (b, c) \in R$. . . . (I)

Then R is transitive

$\Rightarrow (\forall x)(\forall y)(\forall z)((x, y) \in R \wedge (y, z) \in R \Rightarrow (x, z) \in R)$ (Definition of transitivity)

$\Rightarrow ((a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R)$ (UI) . . . (II)

$\Rightarrow (a, c) \in R$ ((I),(II),I3) □

Lemma 5.3.1: *If R is an equivalence relation in X, then the collection of equivalence classes $[X]/R$ is a partition of X.*

Proof: (i) We are to prove that $\bigcup_{S \in [X]/R} S = X$.

(Bidirectional proof)

$\Rightarrow$) $x \in \bigcup_{S \in [X]/R} S \Rightarrow (\exists S)(S \in [X]/R \wedge x \in S)$ $\left(\text{Definition of } \bigcup\right)$

$\Rightarrow A \in [X]/R \wedge x \in A$ (EI)

$\Rightarrow (\exists y)(y \in X \wedge A = [y]/R) \wedge x \in A$ (Definition of $[X]/R$)

$\Rightarrow (\exists y)(y \in X \wedge A = [y]/R)$ and $x \in A$ (E9, I2)

$\Rightarrow (a \in X \wedge A = [a]/R)$ and $x \in A$ (EI)

$\Rightarrow A = [a]/R$ and $x \in A$ (E9, I2)

$\Rightarrow x \in [a]/R$ (Sub$_=$)

$\Rightarrow x \in X \wedge (a, x) \in R$ (Definition of $[a]/R$)

$\Rightarrow x \in X$ (I2)

$\Leftarrow$) Let $x \in X$. . . . (I)

Then $(x, x) \in R$ (R is reflexive)

$\Rightarrow x \in [x]/R$ (Definition of $[x]/R$). . . . (II)

Now, $[x]/R = [x]/R$ (Theorem 3.2.1(i)) . . . (III)

$\Rightarrow (x \in X) \wedge ([x]/R = [x]/R)$ ((I),(III), I6)

$\Rightarrow (\exists y)(y \in X) \wedge ([x]/R = [y]/R)$ (EQ)

$\Rightarrow [x]/R \in [X]/R$ (Definition of $[X]/R$) . . . (IV)

$\Rightarrow ([x]/R \in [X]/R) \wedge (x \in [x]/R)$ ((IV),(II), I6)

$\Rightarrow (\exists S)((S \in [X]/R) \wedge (x \in S))$ (EQ)

$\Rightarrow x \in \bigcup_{S \in [X]/R} S$ (Definition of $\bigcup$)

Therefore, $x \in \bigcup_{S \in [X]/R} S \Leftrightarrow x \in X$.

$\Rightarrow (\forall x)\left(x \in \bigcup_{S \in [X]/R} S \Leftrightarrow x \in X\right)$ (Gen).

$\Rightarrow \bigcup_{S \in [X]/R} S = X$. (Principle of Extension)

(ii) Let $S \in [X]/R$. We are to prove that $S \neq \emptyset$.

$S \in [X]/R \Rightarrow (\exists x)(x \in X \wedge S = [x]/R)$. (Definition of $[X]/R$)

$\Rightarrow (a \in X \wedge S = [a]/R)$ (EI)

$\Rightarrow a \in X$ and $S = [a]/R$ (E9,I2) . . . (I)

$\Rightarrow (a, a) \in R$ ((I), R is reflexive)

$\Rightarrow a \in [a]/R$ (Definition of $[a]/R$)

$\Rightarrow (\exists x) x \in [a]/R$ (EQ)

$\Rightarrow [a]/R \neq \emptyset$ (Lemma 3.5.8)

$\Rightarrow S \neq \emptyset$ ((I), Sub$_=$)

(iii) (Indirect proof)

Let $S, S' \in [X]/R$ such that $S \cap S' \neq \emptyset$. We are to prove that $S = S'$.

Since $S, S' \in [X]/R$,

$(\exists v)(v \in X \wedge S = [v]/R)$ and $(\exists w)(w \in X \wedge S' = [w]/R)$ (Definition of $[X]/R$)

$\Rightarrow (x \in X \wedge S = [x]/R)$ and $(y \in X \wedge S' = [y]/R)$ (EI)

$\Rightarrow x \in X$ and $S = [x]/R$ and $y \in X$ and $S' = [y]/R$ (E9, I2)

First, we want to prove that $(x, y), (y, x) \in R$.

$S \cap S' \neq \emptyset \Rightarrow ([x]/R \cap [y]/R) \neq \emptyset$ (Sub$_=$)

$\Rightarrow (\exists z) z \in ([x]/R \cap [y]/R)$ (Lemma 3.5.8)

$\Rightarrow a \in ([x]/R \cap [y]/R)$ (EI)

$\Rightarrow a \in [x]/R \wedge a \in [y]/R$ (Definition of $\cap$)

$\Rightarrow (x, a) \in R \wedge (y, a) \in R$ (Definition of $[x]/R, [y]/R$)

$\Rightarrow (x, a) \in R \wedge (a, y) \in R$ (R is symmetric)

$\Rightarrow (x, y) \in R$ (R is transitive) . . . (I)

$\Rightarrow (y, x) \in R$ (R is symmetric) . . . (II)

Now, $u \in S \Leftrightarrow u \in [x]/R$ ($S = [x]/R$, Sub$_=$)

$\Leftrightarrow (x, u) \in R$ (Definition of $[x]/R$) . . . (III)

$\Leftrightarrow (y, u) \in R$ ($\Rightarrow$)(II),(III),R is transtive; $\Leftarrow$)(I),(IV),R is transitive) . . . (IV)

$\Leftrightarrow u \in [y]/R$ (Definition of $[y]/R$)

$\Leftrightarrow u \in S'$ ($S' = [y]/R$, Sub$_=$)

We thus have: $(\forall u)(u \in S \Leftrightarrow u \in S')$ (Gen)

Hence, $S = S'$. (Principle of Extension)

In conclusion, $[X]/R$ is a partition of X. ■

Definition: Let $\mathcal{C}$ be a partition of X. The ***relation induced by*** $\mathcal{C}$, denoted by $X/\mathcal{C}$, is a relation in X, such that

$$\begin{aligned} X/\mathcal{C} &= \{(x, y) \in X \times X \mid (\exists S \in \mathcal{C})(x \in S \wedge y \in S)\} \\ &= \{(x, y) \mid (x \in X \wedge y \in X) \wedge (\exists S \in \mathcal{C})(x \in S \wedge y \in S)\} \end{aligned}$$

Since $(x \in X \wedge y \in X) \wedge (\exists S \in \mathcal{C})(x \in S \wedge y \in S) \Leftrightarrow (\exists S \in \mathcal{C})(x \in S \wedge y \in S)$ (left as an exercise), $X/\mathcal{C}$ can thus be redefined as follows:

$$X/\mathcal{C} = \{(x, y) \mid (\exists S \in \mathcal{C})(x \in S \wedge y \in S)\}$$

Example 5.3.9 Consider the equivalence relations in Example 5.3.5. We have shown in Example 5.3.6 that the collection of equivalence classes of each of them is a partition. We shall investigate the relation induced by each of these partitions.

(i) $(x, y) \in X/([X]/(X \times X)) \Leftrightarrow (x, y) \in X/\{X\}$ ($\because [X]/(X \times X) = \{X\}$)

$\Leftrightarrow (\exists S \in \{X\})(x \in S \wedge y \in S)$ (Definition of $X/\mathcal{C}$)

$\Leftrightarrow x \in A \wedge y \in A$, where $A \in \{X\}$ ($\Rightarrow$)EI; $\Leftarrow$)EQ)

$\Leftrightarrow x \in X \wedge y \in X$ ($A \in \{X\} \Leftrightarrow A = X$; Sub$_=$)

$\Leftrightarrow (x, y) \in X \times X$ (Definition of $\times$)

Therefore, $X/([X]/(X \times X)) = X \times X$. (Principle of Extension)

i.e. The relation induced by the collection of equivalence classes w.r.t. $X \times X$ turns out to be the equivalence relation, $X \times X$, itself.

(ii) $(x, y) \in X/([X]/=) \Leftrightarrow (\exists S \in [X]/=)(x \in S \wedge y \in S)$ (Definition of X/C)

$\Leftrightarrow x \in A \wedge y \in A$, *where* $A \in [X]/=$ ($\Rightarrow$)EI; $\Leftarrow$)EQ)

$\Leftrightarrow x \in \{z\} \wedge y \in \{z\}$, *where* $z \in X$ (Definition of $[X]/=$)

$\Leftrightarrow (x = z) \wedge (y = z)$ (Definition of $\{z\} = \{u \mid u = z\}$)

$\Leftrightarrow x = y$ ($\Rightarrow$) = is symmetric, transitive; $\Leftarrow$) let $y = z$, I6, $\text{Sub}_=$)

$\Leftrightarrow (x, y) \in =$ (Definition of =)

Therefore, $X/([X]/=) = =$. (Principle of Extension)

(iii) $(x, y) \in \mathbf{Z}/\left([\mathbf{Z}]/\equiv_m\right) \Rightarrow \left(\exists S \in [\mathbf{Z}]/\equiv_m\right)(x \in S \wedge y \in S)$ (Definition of X/C)

$\Rightarrow x \in A \wedge y \in A$, *where* $A \in [\mathbf{Z}]/\equiv_m$ (EI)

$\Rightarrow x \in [k]/\equiv_m \wedge y \in [k]/\equiv_m$, where $0 \leq k < m$ ($\because$ Example 5.3.7(iii))

$\Rightarrow (\exists q \in \mathbf{Z})(x = qm + k) \wedge (\exists q' \in \mathbf{Z})(y = q'm + k)$, where $0 \leq k < m$.

(Example 5.3.6(iii), note that

$r_k = k$ because $0 \leq k < m$)

$\Rightarrow (x = am + k) \wedge (y = bm + k)$, where $a, b \in \mathbf{Z}$ (EI)

$\Rightarrow x - y = (a - b)m$

$\Rightarrow (\exists Q \in \mathbf{Z})(x - y = Qm)$ ($(a - b) \in \mathbf{Z}$)

$\Rightarrow (x, y) \in \equiv_m$. (Definition of $\equiv_m$)

On the other hand, let $(x, y) \in \equiv_m$. Then $y \in [x]/\equiv_m$. (Definition of $[x]/\equiv_m$)

As $\equiv_m$ is reflexive, $(x, x) \in \equiv_m$ implies $x \in [x]/\equiv_m$. (Definition of $[x]/\equiv_m$)

Therefore, $x, y \in [x]/\equiv_m$ which implies that $\exists S \in [\mathbf{Z}]/\equiv_m$ such that $x \in S \wedge y \in S$.

As a result, $(x, y) \in \mathbf{Z}/([\mathbf{Z}]/\equiv)_m$. (Definition of X/C)

Hence, $\mathbf{Z}/\left([\mathbf{Z}]/\equiv_m\right) = \equiv_m$. (Principle of Extension) ■

In each of the above examples, the relation induced by the collection of equivalence classes of an equivalence relation is the equivalence relation itself. The following lemma shows that this is always the case.

Lemma 5.3.2: *let R be an equivalence relation in X. The relation induced by the collection of equivalence classes* $[X]/R$ *is identical to R. In other words,* $X/([X]/R) = R$

Proof: We want to prove that $(x, y) \in X/([X]/R) \Leftrightarrow (x, y) \in R$.

(Bidirectional proof)

$\Rightarrow$) Let $(x, y) \in X/([X]/R)$.

Then, $(\exists S)(S \in [X]/R \wedge x \in S \wedge y \in S)$ (Definition of X/C)

$\Rightarrow A \in [X]/R \wedge x \in A \wedge y \in A$ (EI)

$\Rightarrow A \in [X]/R$ and $x \in A \wedge y \in A$ (E9, I2)

$\Rightarrow (\exists z)(z \in X \wedge A = [z]/R)$ and $x \in A \wedge y \in A$ (Definition of $[X]/R$)

$\Rightarrow (a \in X \wedge A = [a]/R)$ and $x \in A \wedge y \in A$ (EI)

$\Rightarrow A = [a]/R$ and $x \in A \wedge y \in A$ (E9, I2)

$\Rightarrow (x \in [a]/R) \wedge (y \in [a]/R)$ ($\text{Sub}_{=}$)

$\Rightarrow (a, x) \in R \wedge (a, y) \in R$ (Definition of $[a]/R$)

$\Rightarrow (a, x) \in R$ and $(a, y) \in R$ (E9, I2)

$\Rightarrow (x, a) \in R$ and $(a, y) \in R$ (R is symmetric)

$\Rightarrow (x, a) \in R \wedge (a, y) \in R$ (I6)

$\Rightarrow (x, y) \in R$ (R is transitive)

$\Leftarrow$) Let $(x, y) \in R$. . . . (I)

Then $y \in [x]/R$ (Definition of $[x]/R$) . . . (II)

We shall prove that $x \in [x]/R$.

R is a relation in $X \Rightarrow R \subseteq X \times X$ (Definition of relation)

$\Rightarrow (\forall z)(z \in R \Rightarrow z \in X \times X)$ (Definition of $\subseteq$)

$\Rightarrow ((x, y) \in R \Rightarrow (x, y) \in X \times X)$ (UI) . . . (III)

$\Rightarrow (x, y) \in X \times X$ ((I),(III), I3)

$\Rightarrow x \in X \wedge y \in X$ (Definition of $X \times X$)

$\Rightarrow x \in X$ (I2) . . . (IV)

$\Rightarrow (x, x) \in R$ (R is reflexive)

$\Rightarrow x \in [x]/R$ (Definition of $[x]/R$) . . . (V)

Now, $[x]/R = [x]/R$ (Reflexivity of =) . . . (VI)

$\Rightarrow (x \in X) \wedge ([x]/R = [x]/R)$ ((IV),(VI), I6)

$\Rightarrow (\exists z)(z \in X \wedge ([x]/R = [z]/R))$ (EQ)

$\Rightarrow [x]/R \in [X]/R$ (Definition of $[X]/R$) . . . (VII)

It follows that $[x]/R \in [X]/R \wedge x \in [x]/R \wedge y \in [x]/R$ ((VII)(V)(II), I6)

$\Rightarrow (\exists S)(S \in [X]/R \wedge x \in S \wedge y \in S)$ (EQ)

$\Rightarrow (x, y) \in X/([X]/R)$. (Definition of $X/\mathcal{C}$)

Hence, $(x, y) \in X/([X]/R) \Leftrightarrow (x, y) \in R$

$\Rightarrow (\forall u)(u \in X/([X]/R) \Leftrightarrow u \in R)$ (Gen)

$\Rightarrow X/([X]/R) = R$ (Principle of Extension) ■

So far, we have seen that if $\mathcal{C}$ is a partition which is the collection of the equivalence classes of an equivalence relation in X, then $X/\mathcal{C}$ is an equivalence relation. Does this still hold true if $\mathcal{C}$ is any partition of X? The following lemma gives an affirmative answer.

Lemma 5.3.3: *Let $\mathcal{C}$ be a partition of X. The induced relation $X/\mathcal{C}$ is an equivalence relation in X.*

Proof:

(i) Let $x \in X$. Then $x \in \bigcup_{S \in \mathcal{C}} S$ $(\bigcup_{S \in \mathcal{C}} S = X)$

$\Rightarrow (\exists S)(S \in \mathcal{C} \wedge x \in S)$ (Definition of $\bigcup$)

$\Rightarrow (\exists S)(S \in \mathcal{C} \wedge x \in S \wedge x \in S)$ (E3)

$\Rightarrow (x, x) \in X/\mathcal{C}$ (Definition of $X/\mathcal{C}$)

Therefore, $X/\mathcal{C}$ is reflexive.

(ii) Let $(x, y) \in X/C$. Then $(\exists S \in C)(x \in S \wedge y \in S)$ (Definition of X/C)

$\Rightarrow (\exists S \in C)(y \in S \wedge x \in S)$ (E9)

$\Rightarrow (y, x) \in X/C$ (Definition of X/C)

Therefore, X/C is symmetric.

(iii) Let $(x, y) \in X/C$ and $(y, z) \in X/C$.

Then $(\exists S \in C)(x \in S \wedge y \in S)$ and $(\exists S \in C)(y \in S \wedge z \in S)$ (Definition of X/C)

$\Rightarrow (A \in C \wedge x \in A \wedge y \in A)$ and $(A' \in C \wedge y \in A' \wedge z \in A')$ (EI)

$\Rightarrow A \in C$ and $x \in A$ and $y \in A$ and $A' \in C$ and $y \in A'$ and $z \in A'$ (E9, I2) . . . (I)

$\Rightarrow A \in C \wedge A' \in C$ and $y \in A \wedge y \in A'$ (I6) . . . (II)

$\Rightarrow y \in (A \cap A')$ (Definition of $\cap$)

$\Rightarrow (\exists u) u \in (A \cap A')$ (EQ)

$\Rightarrow (A \cap A') \neq \emptyset$ (Lemma 3.5.8) . . . (III)

C is a partition of $X \Rightarrow \forall S, S' \in C, S \neq S' \Rightarrow (S \cap S') = \emptyset$

$\Rightarrow A \in C \wedge A' \in C \Rightarrow (A \neq A' \Rightarrow (A \cap A') = \emptyset)$ (UI) . . . (IV)

$\Rightarrow (A \neq A' \Rightarrow (A \cap A') = \emptyset)$ ((II),(IV),I3) . . . (V)

$\Rightarrow A = A'$ ((III),(V), I4, E15) . . . (VI)

We thus have $A \in C \wedge x \in A \wedge z \in A'$ ((I), I6)

$\Rightarrow A \in C \wedge x \in A \wedge z \in A$ ((VI), Sub$_=$)

$\Rightarrow (\exists S)(S \in C \wedge x \in S \wedge z \in S)$ (EQ)

$\Rightarrow (x, z) \in X/C$ (Definition of X/C)

Hence, X/C is transitive. ■

Given a partition C of X. Since X/C is an equivalence relation in X, by Lemma 5.3.1, $[X]/(X/C)$ is a partition of X. In the following, we shall show that this partition of X is identical to C. First, let us prove the following lemma.

Lemma 5.3.4: *Let C be a partition of X. Then $A \in C \wedge a \in A \Rightarrow A = [a]/(X/C)$.*

Proof: (Direct proof) Suppose $A \in C \wedge a \in A$.

$$\text{Then } \begin{cases} A \in C & \cdots (\mathbf{I}) \\ a \in A & \cdots (\mathbf{I}') \end{cases} \text{ (E9,I2)}$$

We want to prove that $y \in A \Leftrightarrow y \in [a]/(X/C)$.

(Bidirectional proof)

$\Leftarrow$) (Direct proof)

$y \in [a]/(X/C) \Rightarrow (a, y) \in X/C$ (Definition of $[a]/(X/C)$)

$\Rightarrow (\exists S')(S' \in C \wedge a \in S' \wedge y \in S')$ (Definition of X/C)

$\Rightarrow (B \in C \wedge a \in B \wedge y \in B)$ (EI)

$$\Rightarrow \begin{cases} B \in C & \cdots (\mathbf{II}) \\ a \in B & \cdots (\mathbf{II}') \\ y \in B & \cdots (\mathbf{II}'') \end{cases} \text{ (E9,I2)}$$

$\Rightarrow A \in C \wedge B \in C$ ((I),(II),I6) . . . **(III)**

and $a \in A \wedge a \in B$ ((I'),(II'),I6) . . . **(IV)**

$\Rightarrow a \in A \cap B$ (Definition of $\cap$)

$\Rightarrow (\exists x)x \in A \cap B$ (EQ)

$\Rightarrow A \cap B \neq \emptyset$ (Lemma 3.5.8) . . . **(V)**

C is a partition $\Rightarrow (\forall S')(\forall S'')(S' \in C \wedge S'' \in C \Rightarrow (S' \cap S'' \neq \emptyset \Rightarrow S' = S''))$

$\Rightarrow A \in C \wedge B \in C \Rightarrow (A \cap B \neq \emptyset \Rightarrow A = B)$ (UI) . . . **(VI)**

$\Rightarrow (A \cap B \neq \emptyset \Rightarrow A = B)$ ((III),(VI),I3) . . . **(VII)**

$\Rightarrow A = B$ ((V),(VII),I3) . . . **(VIII)**

$\Rightarrow y \in A.$ ((II"),(VIII), $\text{Sub}_=$)

$\Rightarrow$) (Direct proof) Let $y \in A$. Then $A \in C \wedge a \in A \wedge y \in A$ ((I),(I'), I6)

$\Rightarrow (\exists S')(S' \in C \wedge a \in S' \wedge y \in S')$ (EQ)

$\Rightarrow (a, y) \in X/C$ (Definition of (X/C))

$\Rightarrow y \in [a]/(X/C)$ (Definition of $[a]/(X/C)$)

We thus have $y \in [a]/(X/C) \Leftrightarrow y \in A$

$\Rightarrow (\forall y)(y \in [a]/(X/C) \Leftrightarrow y \in A)$ (Gen)

$\Rightarrow [a]/(X/C) = A$ (Principle of Extension) ■

Lemma 5.3.5: *Let C be a partition of X. Then $[X]/(X/C) = C$.*

Proof: We want to prove that $S \in [X]/(X/C) \Leftrightarrow S \in C$.

(Bidirectional proof)

$\Rightarrow$) (Direct proof) Let $S \in [X]/(X/C)$.

Then $(\exists y)(y \in X \wedge S = [y]/(X/C))$ (Definition of $[X]/(X/C)$)

$\Rightarrow (a \in X \wedge S = [a]/(X/C))$ (EI)

$\Rightarrow a \in X$ and $(S = [a]/(X/C))$ (E9,I2) . . . **(I)**

$a \in X \Rightarrow a \in \bigcup_{S' \in C} S'$ (Definition of Partition (i): $X = \bigcup_{S' \in C} S'$, $\text{Sub}_=$)

$\Rightarrow (\exists S')(S' \in C \wedge a \in S')$ (Definition of $\bigcup$)

$\Rightarrow (A \in C \wedge a \in A)$ (EI) . . . **(II)**

$\Rightarrow A \in C$ and $a \in A$ (E9,I2) . . . **(III)**

$\Rightarrow A = [a]/(X/C)$ (Lemma 5.3.4) . . . **(IV)**

$\Rightarrow A = S$ ((I),(IV), $\text{Sub}_=$) . . . **(V)**

$\Rightarrow S \in C$ ((III),(V), $\text{Sub}_=$)

$\Leftarrow$) (Direct proof) Let $S \in \mathcal{C}$. . . . **(I)**

Then $(\forall S')(S' \in \mathcal{C} \Rightarrow S' \neq \emptyset)$ (Definition of partition (ii))

$\Rightarrow (S \in \mathcal{C} \Rightarrow S \neq \emptyset)$ (UI) . . . **(II)**

$\Rightarrow S \neq \emptyset$ ((I),(II), I3)

$\Rightarrow (\exists x) x \in S$ (Lemma 3.5.8)

$\Rightarrow a \in S$ (EI) . . . **(III)**

$\Rightarrow S \in \mathcal{C} \wedge a \in S$ ((I),(III), I6) . . . **(IV)**

$\Rightarrow S = [a]/(X/\mathcal{C})$ (Lemma 5.3.4) . . . **(V)**

Now, $(S \in \mathcal{C} \wedge a \in S)$ (from (IV))

$\Rightarrow (\exists S')(S' \in \mathcal{C} \wedge a \in S')$ (EQ)

$\Rightarrow a \in \bigcup_{S' \in \mathcal{C}} S'$ (Definition of $\bigcup$)

$\Rightarrow a \in X$ (Definition of Partition (i): $\bigcup_{S' \in \mathcal{C}} S' = X$, $\text{Sub}_=$) . . . **(VI)**

It follows that $a \in X \wedge (S = [a]/(X/\mathcal{C}))$ ((VI),(V), I6)

$\Rightarrow (\exists z)(z \in X \wedge (S = [z]/(X/\mathcal{C})))$ (EQ)

$\Rightarrow S \in [X]/(X/\mathcal{C})$ (Definition of $[X]/(X/\mathcal{C})$)

We thus have $S \in [X]/(X/\mathcal{C}) \Leftrightarrow S \in \mathcal{C}$

$\Rightarrow (\forall S)(S \in [X]/(X/\mathcal{C}) \Leftrightarrow S \in \mathcal{C})$ (Gen)

$\Rightarrow [X]/(X/\mathcal{C}) = \mathcal{C}$ (Principle of Extension) ■

Combining the above four lemmas, we have the following theorem.

Theorem 5.3.6:

(i) If R is an equivalence relation in X, then the set of equivalence classes $[X]/R$ is a partition of X that induces the relation R, and

(ii) if $\mathcal{C}$ is a partition of X, then the induced relation $X/\mathcal{C}$ is an equivalence relation in X whose set of equivalence classes is identical to $\mathcal{C}$.

Theorem 5.3.6(ii) provides us with an alternative way of proving that a relation is an equivalence relation. Specifically, instead of proving that a relation is reflexive, symmetric and transitive, we may prove that it is the relation induced by a partition of the set in which it is defined. The following is an example.

Example 5.3.10 Let A be the set of all students in a classroom and R be a relation in A such that $(a, b) \in R$ iff a and b sit in the same row in the classroom. Show that R is an equivalence relation by using Theorem 5.3.6.

Solution: Let $\mathcal{C} = \{S \subseteq A \mid S \textit{ consists of all the students sitting in a non–empty row}\}$.

(i) We want to prove that $\bigcup_{S \in \mathcal{C}} S = A$.

$x \in \bigcup_{S \in C} S \Rightarrow (\exists S \in C) x \in S$ (Definition of $\bigcup$)

$\Rightarrow Y \in C \wedge x \in Y$ (EI)

$\Rightarrow x \in Y$ and $Y \in C$ (E9, I2)

$\Rightarrow x \in Y$ and $Y \subseteq A$ (Definition of C)

$\Rightarrow x \in Y$ and $(\forall x)(x \in Y \Rightarrow x \in A)$ (Definition of $\subseteq$)

$\Rightarrow x \in Y$ and $x \in Y \Rightarrow x \in A$ (UI)

$\Rightarrow x \in A$ (I3)

On the other hand,

$x \in A \Rightarrow x \in S, for\ some\ S \in C$ (Every student x in the classroom sits in one row)

$\Rightarrow x \in \bigcup_{S \in C} S.$ (Definition of $\bigcup$)

Hence, $\left(x \in \bigcup_{S \in C} S \Leftrightarrow x \in A\right) \Rightarrow (\forall x)\left(x \in \bigcup_{S \in C} S \Leftrightarrow x \in A\right)$ (Gen)

$\Rightarrow \bigcup_{S \in C} S = A$ (Principle of Extension)

(ii) We want to prove that $\forall S \in C, S \neq \emptyset$.

Let $S \in C$. From the definition of C, S consists of the students sitting in a "non-empty" row.

Therefore $S \neq \emptyset$. Hence, $\forall S \in C, S \neq \emptyset$.

(iii) We want to prove that $(\forall S, S' \in C)(S \neq S' \Rightarrow S \cap S' = \emptyset)$.

Let $S, S' \in C$ such that $S \neq S'$. Since every student can sit in exactly one row, it follows that $S \cap S' = \emptyset$.

Hence, C is a partition of A.

Consequently, $X/C = \{(a,b) \mid (\exists S \in C)(a \in S \wedge b \in S)\}$ (Definition of X/C)

$= \{(a,b) \mid a$ and b sit in the same row$\}$. (Definition of C)

$= R.$ (Definition of R)

By Theorem 5.3.6(ii), R is an equivalence relation. ■

5.4 Partial Orders

Definition: A relation R in a set X is a ***partial order*** if R is reflexive, antisymmetric and transitive.

Example 5.4.11 From Example 5.2.4(v), (iv) and (vii), we immediately obtain the following results:

1. Let **R** be the set of real numbers. The relation $\leq$ in **R** is a partial order.
2. Let X be any set. The relation $\subseteq$ in $\mathcal{P}(X)$ is a partial order.
3. The divisibility relation $|$ is a partial order in **N**.

The symbol $\precsim$ is customary used to represent partial orders, and $(a,b) \in \precsim$ is customary written as $a \precsim b$. The notation $a \not\precsim b$ represents $\sim (a \precsim b)$. Notice that the symbol $\precsim$ closely resembles the $\leq$ relation in real numbers which is perhaps the most well-known partial order.

Definition: A ***partially ordered set*** is an ordered pair $(X, \precsim)$ in which X is a set and $\precsim$ is a partial order in X. If $\forall x, y \in X, (x \precsim y) \vee (y \precsim x)$, then $\precsim$ is called a ***total order*** and $(X, \precsim)$ is a ***totally ordered set*** or a ***chain***.

Example 5.4.12

(i) Let $\mathbf{R}$ be the set of real numbers. Then $(\mathbf{R}, \leq)$ is a partially ordered set. Moreover, as for any $x, y \in \mathbf{R}$, either $x \leq y$ or $y \leq x$, $(\mathbf{R}, \leq)$ is a totally ordered set.

(ii) Let X be a set. Then $(\mathcal{P}(X), \subseteq)$ is a partially ordered set. $(\mathcal{P}(X), \subseteq)$ is a totally ordered set iff $X = \emptyset$ or X is a singleton.

(iii) Let $\mathbf{N}$ be the set of positive integers. Then $(\mathbf{N}, |)$ is a partially ordered set. It is not totally ordered because neither $3 \mid 5$ nor $5 \mid 3$. ■

Remark: When the partial order is understood, we normally omit it and simply say that X is a partially ordered set. For instance, for the set of real numbers, it is common to say that $\mathbf{R}$ is a partially ordered set, instead of $(\mathbf{R}, \leq)$ is a partially ordered set.

Definition: A relation R in a set X is a ***strict order*** if R is asymmetric and transitive.

Example 5.4.13

(i) The '*strictly less than*' relation (i.e. $<$) in $\mathbf{R}$ is a strict order in $\mathbf{R}$.

(ii) The '*proper subset of*' relation $\subset$ in the power set $\mathcal{P}(X)$ is a strict order.

The symbol $\prec$ is customary used to represent strict orders, and $(a, b) \in \prec$ is customary written as $a \prec b$. The notation $a \not\prec b$ represents $\sim (a \prec b)$. Notice that the symbol $\prec$ closely resembles the $<$ relation in real numbers which is perhaps the most well-known strict order.

Lemma 5.4.1: *Let $\precsim$ be a partial order in X, and $X_=$ be the relation of equality in X. The relation $\precsim - X_=$ is a strict order in X.*

Proof: We are to prove that $\precsim - X_=$ is (i) asymmetric, and (ii) transitive.

(i) (Proof by contradiction)

Suppose $\precsim - X_=$ is not asymmetric.

Then $\sim (\forall x, y \in X) \sim ((x, y) \in \precsim - X_= \wedge (y, x) \in \precsim - X_=)$ (Definition of asymmetricity)

$\Rightarrow (\exists x, y \in X)((x, y) \in \precsim - X_= \wedge (y, x) \in \precsim - X_=)$ (FE7$_\in$, E15)

$\Rightarrow (a, b \in X) \wedge ((a, b) \in \precsim - X_= \wedge (b, a) \in \precsim - X_=)$ (EI)

$\Rightarrow ((a, b) \in \precsim - X_= \wedge (b, a) \in \precsim - X_=)$ (E9,I2)

$\Rightarrow (a \precsim b \wedge a \neq b) \wedge (b \precsim a \wedge b \neq a)$ (Definitions of $-, X_=$)

$\Rightarrow ((a \precsim b \wedge b \precsim a) \wedge a \neq b) \wedge (b \neq a)$ (E9, E11)

$\Rightarrow a \neq b$ and $(a \precsim b \wedge b \precsim a)$ (E9,I2)

$\Rightarrow a \neq b$ and $a = b$ ($\precsim$ is antisymmetric)

$\Rightarrow (a = b) \wedge (a \neq b)$ (I6)

$\Rightarrow false$ (a contradiction!) (E1)

(ii) (Direct proof) $(x, y) \in \precsim - X_= \wedge (y, z) \in \precsim - X_=$

$\Rightarrow (x \precsim y \wedge x \neq y) \wedge (y \precsim z \wedge y \neq z)$ (Definitions of $-, X_=$)

$\Rightarrow (x \precsim y \wedge y \precsim z) \wedge (x \neq y \wedge y \neq z)$ (E9, E11)

$\Rightarrow (x \precsim y \wedge y \precsim z)$ (I2)

$\Rightarrow (x \precsim z)$ ($\precsim$ is transitive) . . . (I)

It remains to prove that $x \neq z$.

(Proof by Contradiction) Suppose $x = z$. . . . (II)

Then $(x, y) \in \precsim - X_= \wedge (y, z) \in \precsim - X_=$

$\Rightarrow (x, y) \in \precsim - X_= \wedge (y, x) \in \precsim - X_=$ ((II), Sub$_=$)

$\Rightarrow (\exists u, v \in X)((u, v) \in \precsim - X_= \wedge (v, u) \in \precsim - X_=)$ (EQ)

$\Rightarrow (\exists u, v \in X) \sim\sim ((u, v) \in \precsim - X_= \wedge (v, u) \in \precsim - X_=)$ (E15)

$\Rightarrow\sim (\forall u, v \in X) \sim ((u, v) \in \precsim - X_= \wedge (v, u) \in \precsim - X_=)$ (FE7$_\in$)

$\Rightarrow \precsim - X_=$ is not asymmetric, which contradicts Part (i)!

Hence $x \neq z$ . . . (III)

$\Rightarrow (x \precsim z) \wedge (x \neq z)$ ((I),(III), I6)

$\Rightarrow ((x, z) \in \precsim) \wedge ((x, z) \notin X_=)$ (Definition of $X_=$)

$\Rightarrow (x, z) \in \precsim - X_=$ (Definition of $-$)

Therefore, $\precsim - X_=$ is transitive. ■

Definition: Let $\precsim$ be a partial order and $X_=$ is the relation of equality in X. Then $\precsim - X_=$ is called the ***strict order corresponding to*** $\precsim$.

Example 5.4.14

(i) The strict order corresponding to the partial order $\leq$ in **R** is the '*strictly less than*' relation (i.e. $<$) in **R**.

(ii) The strict order corresponding to the partial order $\subseteq$ is the '*proper subset of*' relation $\subset$.

Lemma 5.4.2: *Let $\prec$ be a strict order in X, and $X_=$ be the relation of equality in X. The relation $\prec \cup X_=$ is a partial order in X.*

Proof:

(i) (Direct proof) Let $x \in X$.

$x = x \Rightarrow (x = x) \vee (x \prec x)$ (I1)

$\Rightarrow (x \prec x) \vee (x = x)$ (E10)

$\Rightarrow (x, x) \in \prec \vee (x, x) \in X_=$ (Definitions of $\prec$ and $X_=$)

$\Rightarrow (x, x) \in \prec \cup X_=$ (Definitions of $\cup$)

Hence, $x \in X \Rightarrow (x, x) \in \prec \cup X_=$

$\Rightarrow (\forall x)(x \in X \Rightarrow (x, x) \in \prec \cup X_=)$

Therefore, $\prec \cup X_=$ is reflexive.

(ii) (Direct proof) $(x, y) \in \prec \cup X_= \wedge (y, x) \in \prec \cup X_=$

$\Rightarrow ((x, y) \in \prec \vee (x, y) \in X_=) \wedge ((y, x) \in \prec \vee (y, x) \in X_=)$ (definition of $\cup$)

$\Rightarrow (x \prec y \vee x = y) \wedge (y \prec x \vee y = x)$ (definitions of $\prec, X_=$)

$\Rightarrow (x \prec y \vee x = y) \wedge (y \prec x \vee x = y)$ (= is symmetric, I3$_\vee$)

$\Rightarrow (x = y) \vee ((x \prec y) \wedge (y \prec x))$ (E10, E13)

$\Rightarrow ((x \prec y) \wedge (y \prec x)) \vee (x = y)$ (E10)

$\Rightarrow\sim ((x \prec y) \wedge (y \prec x)) \Rightarrow (x = y)$ (E15,E18) . . . (I)

But $\prec$ is a strict order $\Rightarrow \prec$ is asymmetric

$\Rightarrow\sim ((x \prec y) \wedge (y \prec x))$ (Definition of asymmetricity) . . . (II)

$\Rightarrow x = y$ ((II),(I), I3)

Therefore, $\prec \cup X_=$ is antisymmetric.

(iii) (Direct proof) $(x, y) \in \prec \cup X_= \wedge (y, z) \in \prec \cup X_=$

$\Rightarrow ((x, y) \in \prec \vee (x, y) \in X_=) \wedge ((y, z) \in \prec \vee (y, z) \in X_=)$ (definition of $\cup$)

$\Rightarrow (x \prec y \vee x = y) \wedge (y \prec z \vee y = z)$ (definitions of $\prec, X_=$)

$\Rightarrow ((x \prec y \vee x = y) \wedge (y \prec z)) \vee ((x \prec y \vee x = y) \wedge (y = z))$ (E13)

$\Rightarrow ((y \prec z) \wedge (x \prec y \vee x = y)) \vee ((y = z) \wedge (x \prec y \vee x = y))$ (E9)

$\Rightarrow (x \prec y \wedge y \prec z) \vee (x = y \wedge y \prec z) \vee (x \prec y \wedge y = z) \vee (x = y \wedge y = z)$ (E13, E9) . . . (I)

$\prec$ is a strict order $\Rightarrow \prec$ is transitive

$\Rightarrow ((x \prec y \wedge y \prec z) \Rightarrow x \prec z)$ (Definition of transitivity) . . . (II)

$=$ is an equivalence relation $\Rightarrow =$ is transitive

$\Rightarrow ((x = y \wedge y = z) \Rightarrow x = z)$ (Definition of transitivity) . . . (III)

$(x = y \wedge y \prec z) \Rightarrow x = y$ and $y \prec z$ (I2, E9)

$\Rightarrow x \prec z$ ($\text{Sub}_=$) . . . . . (IV)

$(x \prec y \wedge y = z) \Rightarrow x \prec y$ and $y = z$ (I2, E9)

$\Rightarrow x \prec z$ ($\text{Sub}_=$) . . . . . (V)

We thus have:

$(x \prec y \wedge y \prec z) \vee ((x = y \wedge y \prec z) \vee (x \prec y \wedge y = z) \vee (x = y \wedge y = z))$ ((I), E12) . . . (VI)

$\Rightarrow (x \prec z) \vee ((x = y \wedge y \prec z) \vee (x \prec y \wedge y = z) \vee (x = y \wedge y = z))$ ((VI),(II), I3_V)

$\Rightarrow (x = y \wedge y = z) \vee ((x = y \wedge y \prec z) \vee (x \prec y \wedge y = z) \vee (x \prec z))$ (E10, E12) . . . (VII)

$\Rightarrow (x = z) \vee ((x = y \wedge y \prec z) \vee (x \prec y \wedge y = z) \vee (x \prec z))$ ((VII),(III), I3_V)

$\Rightarrow (x = y \wedge y \prec z) \vee ((x = z) \vee (x \prec y \wedge y = z) \vee (x \prec z))$ (E10,E12) . . . (VIII)

$\Rightarrow (x \prec z) \vee (x = z) \vee (x \prec y \wedge y = z) \vee (x \prec z)$ ((VIII),(IV), I3_V)

$\Rightarrow (x \prec y \wedge y = z) \vee ((x = z) \vee (x \prec z) \vee (x \prec z))$ (E10, E12) . . . (IX)

$\Rightarrow (x \prec z) \vee ((x = z) \vee (x \prec z) \vee (x \prec z))$ ((IX),(V), I3_V)

$\Rightarrow ((x \prec z) \vee (x \prec z) \vee (x \prec z)) \vee (x = z)$ (E10,E12)

$\Rightarrow (x \prec z) \vee (x = z)$ (E4)

$\Rightarrow (x, z) \in \prec \vee (x, z) \in X_=$ (equivalent notation)

$\Rightarrow (x, z) \in \prec \cup X_=$ (definition of $\cup$)

Therefore, $\prec \cup X_=$ is transitive. ■

Since $(\prec \cup X_=) - X_= = \prec$ (Exercise 19), $\prec$ is the strict order corresponding to $\prec \cup X_=$. Therefore, every strict order corresponds to a partial order.

Let $\precsim$ be a partial order and $\prec$ be the corresponding strict order. Owing to the transitive properties of $\precsim$ and $\prec$ and Exercise 5.8.11, if $x_1 \overset{\precsim}{\underset{\prec}{}} x_2$, $x_2 \overset{\precsim}{\underset{\prec}{}} x_3$ and . . . and $x_{n-1} \overset{\precsim}{\underset{\prec}{}} x_n$, where each $\overset{\precsim}{\underset{\prec}{}}$ is either $\precsim$ or $\prec$, we may write $x_1 \overset{\precsim}{\underset{\prec}{}} x_2 \overset{\precsim}{\underset{\prec}{}} x_3 \overset{\precsim}{\underset{\prec}{}} \cdots \overset{\precsim}{\underset{\prec}{}} x_n$. For instance, if $x_1 \precsim x_2$, $x_2 \prec x_3$, $x_3 \precsim x_4$, $x_4 \prec x_5$ and $x_5 \prec x_6$, we may write $x_1 \precsim x_2 \prec x_3 \precsim x_4 \prec x_5 \prec x_6$.

Definition: Let $(X, \precsim)$ be a partially ordered set. An element $a \in X$ is called the ***smallest element*** or ***least element*** of X if $\forall x \in X, a \precsim x$. An element $a \in X$ is called the ***largest element*** or ***greatest element*** of X if $\forall x \in X, x \precsim a$. An element $a \in X$ is called a ***minimal element*** of X if $\sim (\exists x \in X) x \prec a$

where $\prec$ is the strict order corresponding to $\precsim$. An element $a \in X$ is called a ***maximal element*** of X if $\sim (\exists x \in X) a \prec x$.

Example 5.4.15

(i) In the partially ordered set $(\{2, 3, 5, 30, 60, 120, 180, 360, 720\}, |)$, the largest element is 720. This is *not* because it is larger than (in the sense of >) every other integer in the set but because it is divisible by every other integer in the set. There is no smallest element in the set because no integer in the set divides every integer in the set. There is only one maximal element which happens to be 720 because it is the only integer which divides no other integer in the set. There are three minimal elements, namely 2, 3 and 5. This is because none of them is divisible by any other integer in the set.

(ii) Let $\mathbf{N}$ be the set of positive integers. The partially ordered set $(\mathbf{N}, \leq)$ has no largest element and no maximal element. However, it has one minimal element which is the integer 1, which also happens to be the smallest element.

(iii) Let $\mathbf{R}$ be the set of real numbers and $(0, 1) = \{x \in \mathbf{R} \mid 0 < x < 1\}$. The partially ordered set $((0, 1), \leq)$ has no smallest element, no minimal element, no largest element and no maximal element. ■

Lemma 5.4.3: The smallest (largest, respectively) element is unique, if exists.

Proof: Suppose a and a' are two smallest elements of a partially ordered set X.

Then a is a smallest element $\Rightarrow (\forall x)(a \precsim x)$ (definition of *smallest element*)

$\Rightarrow a \precsim a'$ (UI)

a' is a smallest element $\Rightarrow (\forall x)(a' \precsim x)$ (definition of *smallest element*)

$\Rightarrow a' \precsim a$ (UI)

But then $a \precsim a'$ and $a' \precsim a$ implies that $a = a'$ ($\because \precsim$ is antisymmetric)

Therefore, the smallest element is unique. The uniqueness of the largest element can be proved similarly. ■

Remarks:

1. The minimal element (maximal, respectively) is non-unique. For instance, in Example 5.4.15(i), the partially ordered set has three minimal elements.

2. Every smallest (largest, respectively) element is a minimal (maximal, respectively) element. The converse is not true.

3. For the set of real numbers, we are accustomed to the property that $x \not< a \Leftrightarrow a \leq x$. This does not always hold true in a partially ordered set unless the set is totally ordered. Otherwise, we would have:

a is a minimal element $\Leftrightarrow \sim (\exists x \in X) x \prec a$ (Definition of minimal element)

$\Leftrightarrow (\forall x \in X) \sim (x \prec a)$ ($\text{FE8}_{\in}$)

$\Leftrightarrow (\forall x \in X)(a \precsim x)$ (Assuming that $\sim (x \prec a) \Leftrightarrow a \precsim x$)

$\Leftrightarrow a$ is the smallest element.

which is clearly false. However, $\sim (x \prec a) \Leftrightarrow (x \precsim a \Rightarrow x = a)$ always holds true as is shown below.